**Errata** Version 15 Jan 2025

This is the errata for the book

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The author's copy with errors corrected can be found in the following link:

`https://xiaoluhou.github.io/Textbook.pdf`

| Location | Original | Change |
|---|---|---|
| Page 8, Theorem 1.1.4 (Euclid's division) | Given $m, n \in \mathbb{Z}$, take $q, r$ such that $n = qm + r$. Then $\gcd(m, n) = \gcd(m, r)$. | Given $m, n \in \mathbb{Z}$, take $q, r \in \mathbb{Z}$ such that $n = qm + r$, and $0 \le r < n$. Then $\gcd(m, n) = \gcd(m, r)$. |
| Page 16, Definition 1.2.8 | The distributive laws holds | The distributive laws hold |
| Page 20, Example 1.2.24 | $f(1 \oplus 0) = f(1) = a,\ f(1) + f(0) = a + b = a$ | $f(1 \oplus 0) = f(1) = b,\ f(1) + f(0) = b + a = b$ |
| Page 9, Algorithm 1.1, lines 2-4 | **Input:** $m,\ n$// $m, n \in \mathbb{Z},\ m \neq 0$<br>**Output:** $\gcd(m, n)$<br>1 **while** $m \neq 0$ **do**<br>2 $r = n\%m$// remainder of $n$ divided by $m$<br>3 $n = m$<br>4 $m = r$<br>5 **return** $r$ | **Input:** $m,\ n$// $m, n \in \mathbb{Z},\ m \neq 0$<br>**Output:** $\gcd(m, n)$<br>1 **while** $m \neq 0$ **do**<br>2 $r = m$<br>3 $m = n\%m$// remainder of $n$ divided by $m$<br>4 $n = r$<br>5 **return** $n$ |
| Page 18, first paragraph below Definition 1.2.12 | By definition, for any $a \in F$, there exists $b \in F$ such that ... | By definition, for any $a \in F$, $a \neq 0$, there exists $b \in F$ such that ... |
| Page 20, Example 1.2.24 | $f(1 \oplus 0) = f(1) = a,\ f(1) + f(0) = a + b = a$ | $f(1 \oplus 0) = f(1) = b,\ f(1) + f(0) = b + a = b$ |
| Page 21, last paragraph | If $a_{ij} = 0$ for $i \neq j$, $A$ is said to be a *diagonal matrix*. An $n-$*dimensional identity matrix*, denoted $I_n$, is a diagonal matrix whose diagonal entries are 1, i.e. $a_{ii} = 1$ for $i = 0, 1, \ldots, n-1$. ... An $n \times n$ matrix is called a *square matrix* (i.e. a matrix with the same number of rows and columns). | An $n \times n$ matrix is called a *square matrix* (i.e. a matrix with the same number of rows and columns). If $A$ is a square matrix and $a_{ij} = 0$ for $i \neq j$, then $A$ is said to be a *diagonal matrix*. An $n \times n$ *identity matrix*, denoted $I_n$, is an $n \times n$ diagonal matrix whose diagonal entries are 1 and all the other entries are 0, i.e. $a_{ii} = 1$ for $i = 0, 1, \ldots, n-1$ and $a_{ij} = 0$ for $i \neq j$. ... |
| Page 44, above Example 1.4.17 | $x = \sum_{i=1}^{3} a_i y_i M_i \bmod n = 2 \times 2 \times 35 + \ldots$ | $x = \sum_{i=1}^{3} a_i y_i M_i \bmod m = 2 \times 2 \times 35 + \ldots$ |
| Page 44, Example 1.4.17 | With the extended Euclidean algorithm, we have found | With the extended Euclidean algorithm, we can find |
| Page 48, Equation (1.23) | $f(x) \times_{F[x]} g(x) := d_n x^n + d_{n-1} x^{n-1} + \ldots$ | $f(x) \times_{F[x]} g(x) := d_{m+n} x^{m+n} + d_{m+n-1} x^{m+n-1} + \ldots$ |
| Page 49, Theorem 1.5.1 | of $\deg(f(x)) \ge 1$ | if $\deg(f(x)) \ge 1$ |
| Page 51, Example 1.5.6 | $\mathbb{F}_2[x]/(f(x)) = \{1, x, x+1\}$<br>...<br>$\mathbb{F}_2[x]/(g(x)) = \{1, x, x+1\}$ | $\mathbb{F}_2[x]/(f(x)) = \{0, 1, x, x+1\}$<br>...<br>$\mathbb{F}_2[x]/(g(x)) = \{0, 1, x, x+1\}$ |
| Page 59, Definition 1.6.6 | A binary code $C$ is said to be $k-$*error correcting* if the minimum distance decoding outputs the correct codeword... | A binary code $C$ is said to be $k-$*error correcting* if with the incomplete decoding rule, minimum distance decoding outputs the correct codeword... |
| Page 105 | $\boldsymbol{c} = c_1 c_2 \cdots = e_k(p_1) e_k(p_2) \ldots.$ | $\boldsymbol{c} = c_1 c_2 \cdots = E_k(p_1) E_k(p_2) \ldots.$ |
| Page 105 | $\boldsymbol{c} = c_1 c_2 \cdots = e_{z_1}(p_1) e_{z_2}(p_2) \ldots$ | $\boldsymbol{c} = E_{z_1}(p_1) E_{z_2}(p_2) \ldots$ |
| Page 106 Table 2.2 (b) | Á 11000001 C1<br>Ä 11000100 C4<br>Í 11001101 CD<br>× 11010111 D7<br>÷ 11110111 F7 | Á 1100001110000001 C381<br>Ä 1100001110000100 C384<br>Í 1100001110001101 C38D<br>× 1100001110010111 C397<br>÷ 1100001110110111 C3B7 |
| Page 107 | A cipher is said to be *computationally secure* if breaking it requires computing power that is not available in practice. | A cipher is said to be *computationally secure* if any attack that can be carried out in realistic (efficient) time has only a negligible chance of breaking it. |
| Page 133 | When $\omega_1 = \omega_2$...the Sbox is a $\omega_1-$bit Sbox | When $\omega_1 = \omega_2$...the Sbox is an $\omega_1-$bit Sbox |
| Page 139, RSA security | Nevertheless, post-quantum public key cryptosystems are being proposed (see e.g. [HPS98, BS08]) to protect communications after a quantum computer is built. | Nevertheless, post-quantum public key cryptosystems are being proposed (see e.g. [HPS98, BS08]) to protect communications after a sufficiently strong quantum computer is built. |
| Page 160, Example 3.2.4 last sentence | Then $\varphi_0(\boldsymbol{x}) = 0$. | Then $\varphi_0(\mathtt{0}) = 0$. |
| Page 168, third line | Let $n = pq$. We choose $e \in \mathbb{Z}^*_{\varphi(n)}$, and compute $d = e^{-1} \bmod \varphi(n)$. | Let $n = pq$. We choose $e \in \mathbb{Z}^*_{\varphi(n)}$, $e \neq 1$, and compute $d = e^{-1} \bmod \varphi(n)$. |
| Page 170, first paragraph | which is computationally infeasible according to property (c) of hash functions listed in Sect. 2.1.1. | which is computationally infeasible according to property (b) of hash functions listed in Sect. 2.1.1. |
| Page 177 | $m = m_p y_q q + m_q y_p p \bmod n = 2 \times 2 \times 5 + 2 \times 2 \times 3 = 32 \bmod 15 = 2$. | $m = m_p y_q q + m_q y_p p \bmod n = 2 \times 2 \times 5 + 2 \times 2 \times 3 \bmod 15 = 32 \bmod 15 = 2$. |
| Page 209, last paragraph of Section 4.1.1 | Similar to SPA, the attack does not require statistical analysis of the traces, only visual inspection is enough. | The sentence should be removed |
| Page 228 | $\frac{\alpha}{2} = 1 - \Phi(z_\alpha) = 1 - \Phi(4.5) = \ldots$ | $\frac{\alpha}{2} = 1 - \Phi(z_{\alpha/2}) = 1 - \Phi(4.5) = \ldots$ |
| Page 236, Example 4.2.15 | $\mathrm{E}\left[\mathrm{wt}\,(\boldsymbol{v})^2\right] = \frac{1}{\lvert\mathbb{F}_2^8\rvert} \sum_{\boldsymbol{v} \in \mathbb{F}_2^8} \mathrm{wt}\,(\boldsymbol{v}^2) = \ldots$ | $\mathrm{E}\left[\mathrm{wt}\,(\boldsymbol{v})^2\right] = \frac{1}{\lvert\mathbb{F}_2^8\rvert} \sum_{\boldsymbol{v} \in \mathbb{F}_2^8} \mathrm{wt}\,(\boldsymbol{v})^2 = \ldots$ |
| Page 248, Remark 4.3.1 | For AES, the correlations between the first AddRoundKey outputs are higher than correlations between the first SubBytes operation outputs, that is why in... | For the PRESENT cipher, correlations among outputs from the initial `addRoundKey` operation are stronger than those between outputs of the initial `sBoxLayer`. Therefore, in... |
| Page 255, Step 8 last sentence | ...when the target signal $\mathrm{wt}(\boldsymbol{v})$. | ...when the target signal is $\mathrm{wt}(\boldsymbol{v})$. |
| Page 255, Step 10 | ...We argue that this is achievable ...is in procession of a clone device | ...We argue that this is achievable ...is in possession of a clone device. |
| Page 262, last sentence | With our profiling traces, we can compute $M_{signal}$ templates. | With our profiling traces, we can compute $M_{signal}$ templates, with each template correspond to one possible value of the target signal. |
| Page 263, first paragraph | For our illustrations, when the target signal is $\boldsymbol{v}$, we will have 16 templates. And when the target signal is $\mathrm{wt}\,(\boldsymbol{v})$, we will have 5 templates. | For our illustrations, when the target signal is $\boldsymbol{v}$, we obtain 16 templates, each corresponding to a possible value of $\boldsymbol{v}$ from `0` to `F`. When the target signal is $\mathrm{wt}\,(\boldsymbol{v})$, we derive 5 templates, each corresponding to a Hamming weight value from 0 to 4. |
| Page 263 Template Step c | For a fixed key hypothesis $\hat{k}_i$, we divide the $M_p$ attack traces from P-DPA Step 10 into $M_{signal}$ sets, $A_1, A_2, \ldots, A_{M_{signal}}$, depending on the hypothetical target intermediate value $\hat{\boldsymbol{v}}_{ij}$ obtained in P-DPA Step 11. In particular, for an attack trace $\boldsymbol{\ell}_j$, let $s_{ij}$ denote the index of the set that it belongs to. Namely<br>$\boldsymbol{\ell}_j \in A_{s_{ij}}$ given key hypothesis $\hat{k}_i$.<br>We are only interested in the leakages at the POIs for each attack trace $\boldsymbol{\ell}_j = (l_1^j, l_2^j, \ldots, l_q^j)$. Define<br>$\boldsymbol{\ell}_{j,\mathrm{POI}} := (l_{t_1}^j, l_{t_2}^j, \ldots, l_{t_{q_{\mathrm{POI}}}}^j)$.<br>With the mean vector $\boldsymbol{\mu}_{s_{ij}}$ and the covariance matrix $Q_{s_{ij}}$ obtained in Template Step b, we can compute the probability of $\boldsymbol{\ell}_j$ given $\hat{k}_i$ using the PDF ... | We are only interested in the leakages at the POIs for each attack trace $\boldsymbol{\ell}_j = (l_1^j, l_2^j, \ldots, l_q^j)$. Define<br>$\boldsymbol{\ell}_{j,\mathrm{POI}} := (l_{t_1}^j, l_{t_2}^j, \ldots, l_{t_{q_{\mathrm{POI}}}}^j)$.<br>For each key hypothesis $\hat{k}_i$ and attack trace $\boldsymbol{\ell}_j$, we compute the hypothetical target intermediate value given the knowledge of the associated plaintext. Let $\mu_{s_{ij}}$ and $Q_{s_{ij}}$ be the template for this hypothetical value, corresponding to $\hat{k}_i$ and $\boldsymbol{\ell}_j$, as obtained in Template Step b. The probability of $\boldsymbol{\ell}_j$ given $\hat{k}_i$ can then be computed using the PDF ... |
| Page 267 | $\mu_1 = -0.039027, \quad \sigma_1^2 = 2.1679112 \times 10^{-6}$. | $\mu_1 = -0.039027, \quad \sigma_1^2 = 2.16437 \times 10^{-6}$. |
| Page 268, Figure 4.46 caption | The target signal is given by the exact value of the 1st Sbox output. | The target signal is given by the exact value of the 6th Sbox output. |
| Page 339, Figure 4.90 caption | Estimations of guessing entropy computed... | Estimations of success rate computed... |
| Page 334, below Equation (4.83) | The size of T1 is $8 \times 4$, and the storage required is $2^8 \times 2^4 = 2^{12}$ bits, or $2^9$ bytes. The table T2 requires 16 bits of memory. | The size of T1 is $8 \times 4$, and the storage required is $2^8 \times 4 = 2^{10}$ bits, or $2^7$ bytes. The table T2 requires $2^4 \times 4 = 64$ bits of memory. |
| Page 395, the last equation | $q = \gcd(s'^e - m, n) = \gcd(7^{11} - 2143) = \gcd(1977326741, 143)$. | $q = \gcd(s'^e - m, n) = \gcd(7^{11} - 2, 143) = \gcd(1977326741, 143)$. |